Interpretable Multilingual Attribution of Frontier LLMs

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***Abstract*—Most research on telling large language models apart uses English, and the accuracy cost of interpretable features against character n-grams is unmeasured on content-controlled multilingual data. We collected 840 persuasive essays from five frontier models (GPT-5.5, Gemini 3.5 Flash, Claude Opus 4.7, Grok 4.3, DeepSeek V4 Pro) in seven languages using 12 content-matched prompts, computed 21 features identically from Universal Dependencies parses, and evaluated five-way attribution with leave-one-prompt-out cross-validation. Logistic regression reaches .59 to .72 accuracy in every language (chance 20%), and .54 to .76 on prompts whose reading-level instruction it never saw. We detect no decline across a coarse resource ranking fixed in advance (-0.038 log-odds per rank step, 95% CI -0.12 to $+0.04$), but the interval allows a 17-point drop. Character n-grams are 3 to 22 points more accurate within a language; given the same unlabeled adaptation, the features transfer between languages at mean .49 against .41 for n-grams. Five features chosen on the training folds reach 78 to 106% of the full set's macro-F1. Attribution on machine-translated text stays at .57 to .70, while the five models, asked which model wrote each text, score 20.3% to 25.4%. In this setting (one genre, fixed prompts, one sampling temperature), non-English machine text can be attributed with features a person can inspect.**

***Index Terms*—authorship attribution, AI-generated text, multilingual text mining, interpretable features, large language models**

1. Introduction

Most work on machine-generated text asks whether a text was written by a person or by a model. Provenance also asks *which* model wrote it. Can readable features distinguish five frontier models in seven languages? What accuracy do we lose against character n-grams? The model-attribution literature is mostly English [1], [2], and the two multilingual studies we know of use fine-tuned or probability-based classifiers whose decisions cannot be read [3], [4].

A moderator tracing a coordinated campaign in Hindi, or a teacher reviewing a Turkish essay, needs to know which paragraphing and punctuation habits support an attribution. A black-box score gives no such reason, and a fine-tuned detector for each language and model version rarely exists outside English.

English results need not carry over. Models see less training text in lower-resource languages [5], so their writing may be less distinctive. Script and morphology also differ: many stylometric measures are undefined outside English or track language more than authorship. Character n-grams, the single most successful feature in authorship attribution [6], have the opposite problem: they see vocabulary and morphology, so they are tied to the script and lexicon they were fit on.

We hold content fixed across languages, compute features identically from Universal Dependencies (UD) parses [7], and compare their accuracy with an n-gram baseline. An analysis plan committed publicly before data collection (Sec. 3.3; RQ5 and RQ6 were added before their data were collected) asks whether five frontier models can be told apart within each of seven languages (**RQ1**), whether attribution weakens with a language's resource level (**RQ2**), whether a model's style is shared across languages (**RQ3**), whether models converge in lower-resource languages (**RQ4**), whether machine translation erases the signal (**RQ5**) and whether the models themselves can identify the author (**RQ6**). The n-gram baseline, the feature-economy analysis and the robustness checks were added afterwards.

Table 2 answers each question in one line and says when to use which representation.

2. Related Work

Detection and stylometry. Human-versus-machine detectors fine-tuned on English lose about a quarter of their F1 on other languages [8]. For human authors, character n-grams are the most successful single feature [6], masking all but the most frequent words isolates a structural signal [9], and UD-derived syntactic features attribute authors in several languages [10].

Model attribution. In English, attribution among neural generators [1] uses stylometric [11] and embedding classifiers, which separate model families and largely survive LLM translation into Chinese [2]; models optimized to fool detectors keep a distinct style [12]. La Cava et al. attribute

among seven older LLMs and human text in 18 languages with fine-tuned and probability-based detectors and find generators harder to distinguish in English [3]; MultiGhost-Bench attributes long-form text to five recent LLMs in six languages and finds fine-tuned transformers more language-independent than statistical detectors [4]. In Japanese, Zaitsu et al. attribute among six LLMs with part-of-speech and function-word features [13]. To our knowledge, no prior which-model study uses one feature set with the same computational definition in every language, holds content fixed across languages, and compares it with character n-grams under matched adaptation within and across languages.

Homogenization and LLM judges. Writing with LLMs reduces the diversity of human text [14]; whether models converge on each other as training data thins has not been tested. LLM evaluators favor their own outputs [15], and models asked to name the author of a single English text score near chance and default to GPT or Claude [16]; RQ6 repeats that task in seven languages.

3. Data and Method

3.1. Models, Languages and Prompts

We queried the five subject models (GPT-5.5, Gemini 3.5 Flash, Claude Opus 4.7, Grok 4.3, DeepSeek V4 Pro) through OpenRouter in September 2026 and logged each response's served model identifier. The seven languages are English, Spanish, Chinese, Russian, Japanese, Turkish and Hindi, assigned ordinal resource ranks 1 to 7 in that order before data collection as a coarse proxy for their share of web training data. The ranking is contestable: Joshi et al. [5] place English, Spanish, Chinese and Japanese in their highest class (5) and Russian, Turkish and Hindi one lower (4), so we report that two-class split beside the planned order.

Twelve English prompt schemas each fix a topic, a stance (six for, six against), three supporting claims, a prose-only instruction and a reading level (four prompts each for general, educated and expert readers). For each language a model that is not a subject (Llama 4 Maverick) wrote a native prompt from the schema rather than translating the English, so the wording is idiomatic while the content is fixed. A model from a third family (Qwen 3.7 Plus) read each of the 84 prompts blind, extracted topic, stance and claims, and compared them with the schema; failed prompts were regenerated with the reviewer's note until all 84 passed.

3.2. Generation, Validation and Features

We sent each prompt as a single user turn without a system prompt at temperature 0.7, twice per cell with fixed seeds and hidden reasoning excluded, yielding $5 \times 12 \times 7 \times 2 = 840$ responses. Each passed language identification (confidence ≥ 0.6), a 150 to 900 word-equivalent window and a per-language refusal check. One Grok response to a Hindi prompt was written in English and excluded, leaving 839 texts; ten DeepSeek responses that exhausted their token budget on hidden reasoning were re-collected with a larger budget.

We parsed texts with Stanza [17], pre-splitting Chinese and Japanese on sentence-final punctuation. We compute 21 features in five groups. *Lexical*: moving-average type-token ratio (MATTR, window 50), hapax rate, mean token length, Zipf slope. *Syntactic*: mean and SD of sentence length, burstiness of sentence length, mean maximum dependency depth, subordinate-clause rate (`acl`, `advcl`, `ccomp`, `xcomp`, `csubj` per sentence), function-word ratio, first-person rate. *Structure*: paragraph count and length, question rate, connective rate. *Punctuation*: commas, colons, dashes and semicolons per 1,000 tokens, with full-width and script variants mapped to the same feature. *Character*: character-bigram entropy and digit rate.

3.3. Analysis

The analysis plan (`docs/analysis_plan.md`) was committed on 3 September 2026 (commit `82673ac`; the repository and its history are public) before any response was collected. RQ5 was added before any translation existed (`88e08a7`) and RQ6 before the judge data were collected, after a small pilot that set the reasoning condition (`60593f5`, `c7232ff`).

Validation and tests. The two generations of a prompt, and its versions in seven languages, are related, so every split is leave-one-prompt-out (LOPO, 12 folds), every interval is a prompt-clustered bootstrap, the GLMs use prompt-clustered errors, and accuracy comparisons are tested on the 12 per-prompt accuracies rather than on texts: an exact sign-flip permutation test over 4,096 sign assignments (smallest one-sided p = 1/4,096). Holm correction runs over the 28 headline comparisons (features and n-grams vs. chance, features vs. n-grams, features vs. the best judge, in seven languages), over the 42 transfer cells of each representation, and over the five judges. Spearman p-values over seven languages are exact.

Classifiers and tuning. Features are standardized inside each training fold (none has missing values) and fed to L_2-regularized multinomial logistic regression. The baseline fits TF-IDF over character 1–4-grams within word boundaries (minimum document frequency 2, sublinear term frequency, at most 60,000 terms) on the training prompts only, then the same classifier; word 1–2-grams likewise. The features use $C = 1$, fixed in the plan; the later n-gram baseline uses the same C, and its TF-IDF settings were set once, not tuned. As a check, we also choose $C \in \{0.01, 0.1, 1, 10, 100\}$ for both by nested LOPO on the training prompts.

RQ2. A binomial GLM of cell-level correctness on resource rank, clustered by prompt, language, and both. **RQ3.** Per-feature ANOVA (model $\times$ language, prompt as block) with partial η^2, and transfer from language A to B after within-language z-scoring (all texts of a language, no labels). The n-gram counterpart fits the vocabulary on the union of A and B without labels and z-scores TF-IDF columns within each language, so both representations get

TABLE 1. PER-LANGUAGE RESULTS IN RESOURCE-RANK ORDER. UD: LOGISTIC REGRESSION ON THE 21 FEATURES, ACCURACY [PROMPT-CLUSTERED 95% CI] AND MACRO-F1; n-GRAM: CHARACTER 1–4-GRAM TF-IDF [CI]; JUDGES: MEAN OF THE FIVE LLM JUDGES (BEST); SEP.: MEAN CENTROID DISTANCE (z-UNITS). CHANCE IS 0.20.

Language	UD [95% CI]	F1	n-gram [CI]	Judges	Sep.
English	.717 [.61–.82]	.711	.750 [.71–.79]	.230 (.267)	3.59
Spanish	.625 [.53–.70]	.625	.733 [.65–.80]	.207 (.217)	3.11
Chinese	.650 [.55–.73]	.644	.867 [.81–.93]	.208 (.233)	3.01
Russian	.650 [.52–.78]	.654	.783 [.66–.88]	.237 (.258)	2.97
Japanese	.708 [.63–.78]	.707	.792 [.71–.87]	.198 (.225)	2.70
Turkish	.592 [.50–.69]	.596	.783 [.73–.84]	.205 (.258)	2.84
Hindi	.639 [.56–.71]	.634	.782 [.73–.83]	.247 (.361)	2.69

TABLE 2. ANSWERS, AND WHEN TO USE WHICH REPRESENTATION (THIS GENRE, PROMPT SET AND MODEL SNAPSHOT).

Question	Answer
RQ1 Within one language?	Yes: .59–.72; .54–.76 on unseen reading level.
RQ2 Resource gradient?	No decline detected; up to 17 points not excluded.
RQ3 Across languages?	Partly: .37–.59 between language pairs.
RQ4 Convergence?	No by the planned rule; Grok's English style fades.
RQ5 Erased by translation?	No: .57–.70 on machine translations.
RQ6 Can the models tell?	No: 20.3%–25.4%; only Claude is above chance.
Use case	**Better representation**
Target-language labels	Character n-grams (+3 to 22 points).
Explanation needed	Features (five keep 78–106% of macro-F1).
Cross-language use	Features (.49 vs. .41 for n-grams).
Labeled translations	Character n-grams (.63–.78 vs. .57–.70).

the same unlabeled adaptation. **RQ4.** After within-language z-scoring, the mean pairwise distance between the five model centroids, the silhouette of model labels and the between/within scatter ratio. **RQ5.** All 120 English responses were translated into the six other languages by Google Translate and by an open non-subject model (MiniMax M3); we measure attribution on the translations alone, transfer from classifiers trained on the English originals or on native target-language responses, and per-feature Spearman correlations between originals and translations. **RQ6.** Each subject model was asked, one text at a time with a shuffled candidate list and no metadata, which of the five wrote it (temperature 0, hidden reasoning off where the API allows), the exact-model-prediction task of Bai et al. [16]. **Feature economy.** A nested selection curve (features ranked by mean absolute standardized coefficient on the training fold only, top k refit), group ablations, and a character model that sees only punctuation, digits and whitespace [9]. **Fragility.** Leave-one-reading-level-out: train on the eight prompts of two levels, test on the four of the third; and LOPO without the two paragraph features.

4. Results

4.1. Within-Language Attribution

Features distinguish models in every language (Table 1). Accuracy is .59 to .72; every prompt-clustered interval excludes chance, and every prompt-level test against chance has Holm $p \leq 0.007$. The 21 inspectable features identify the author about three times as often as guessing. Length is part of the signal: with features residualized on log length, logistic regression falls to .44 to .57, still far above chance.

n-gram baseline. Character n-grams reach .73 to .87, 3 to 22 points above the features (mean 13). They are ahead in 7 to 11 of the 12 prompts in every language but English (4), yet with 12 prompts the advantage is Holm-significant only in Chinese (next, Turkish: $p = 0.059$). Choosing C by nested LOPO for both sides (selected values 0.01–10 for features, 0.1–100 for n-grams) moves mean accuracy by less than one point for the features and by +2 for n-grams, widening the gap to 8 to 23 points (mean 15); the untuned comparison does not flatter the features. N-grams capture vocabulary and morphology that the features omit [6].

Fragility. Trained on two reading levels and tested on prompts with the third, the features reach .54 to .76 (mean change -3 points from LOPO; n-grams .66 to .88, -5). Without the two paragraph features, which a layout instruction would change most directly, they reach .51 to .69 under LOPO and .52 to .68 across reading levels, every prompt-clustered interval above chance (lowest bound .41). We did not vary format instructions or temperature.

LLM judges (RQ6). Judges identify authors near chance. Given texts individually, they score 20.3% (DeepSeek) to 25.4% (Claude). Only Claude is above chance (all 12 prompts; Holm over five judges $p = 0.001$, next judge $p = 0.096$) and names itself more often for its own texts (own-text recall 17% against 7% false self-claims; 10 of 12 prompts, $p = 0.015$); GPT-5.5 names itself for 82% of texts regardless of author. The feature classifier exceeds the best judge by 28 to 48 points in every language (Holm $p \leq 0.016$). Bai et al. [16] report the same near-chance, GPT- and Claude-biased behavior in English; here it holds in every language except for Claude in Hindi (.361). We did not test side-by-side comparison or extended reasoning.

4.2. Feature Importance

The strongest features are paragraph count, comma rate, char-bigram entropy, paragraph length, Zipf slope, ranked by mean absolute standardized coefficient. In the ANOVA, language explains far more variance than model (mean partial η^2 .54 against .145; interaction .13; prompt .16). Model effects are largest for hapax rate (.38), Zipf slope (.37), comma rate (.32), char-bigram entropy (.31), paragraph count (.29) and smallest for subordination, question rate and dependency depth ($\leq$ .03): models differ in lexical variety, punctuation and paragraphing, hardly at all in clause structure.

A few features retain most attribution performance (Fig. 1a). The five features ranked highest on training folds reach 78 to 106% of the full set's macro-F1 (.50 to .75 against .60 to .71). The ranking changes with language (paragraph count is first in Russian, sixteenth in English); the pooled order is paragraph count, paragraph length, sentence length, comma rate, hapax rate. No single group is required: removing one costs at most .11 macro-F1 (punctuation in Hindi, syntax in Turkish); structure alone gives .31

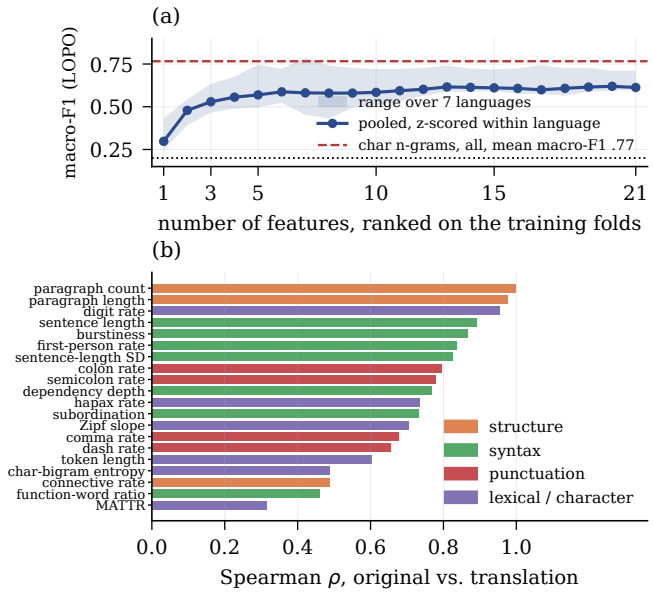


Figure 1. (a) Nested feature-selection curve; features are ranked on the training folds only, and the band is the range over the seven per-language curves. (b) Spearman correlation between each English original and its machine translation, per feature, averaged over six languages and two translators; question rate is omitted because the English essays contain no questions.

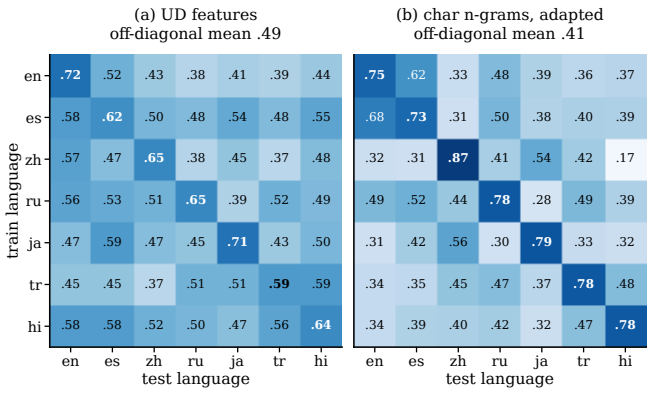


Figure 2. Cross-language transfer: a classifier trained on one language (rows) is tested on another (columns) after within-language z-scoring; the diagonal is LOPO within language. (a) UD features. (b) Character n-grams given the same adaptation (union vocabulary, TF-IDF columns z-scored per language).

to .61 and punctuation alone .26 to .53. Punctuation, digits and whitespace give .49 to .81 accuracy within language: segmentation, punctuation and length carry a signal.

4.3. Transfer and Translation

Cross-language transfer (RQ3). Features transfer above chance between all languages (Fig. 2a). With within-language z-scoring, accuracy is .37 to .59 (mean .49, against .65 within language); all 42 off-diagonal cells exceed chance at the prompt level after Holm correction ($p \leq 0.021$), retaining 63% of above-chance accuracy. With equal unlabeled adaptation, character n-grams average .41 (Fig. 2b) and are significant in 16 of 42 cells; features win in 33 of 42 cells and on all 12 prompts (paired sign-flip $p = 0.0005$). Without adaptation, n-grams transfer at .28. The test language contains the 12 training prompts, so off-diagonal cells are an upper bound for unseen prompts.

Translation (RQ5). Translation preserves attribution signal. Training and testing on translations (LOPO) gives .57 to .70, every prompt-clustered interval above chance (lowest bound .48), about 9 points below English originals (.717) and close to native responses (mean .644); retrained n-grams reach .63 to .78. Paragraph count ($\rho = 1.00$), paragraph length (.98), digit rate (.95) and sentence length (.89) survive largely by construction (Fig. 1b): both translators keep boundaries and copy digits. Comma rate (.68) survives with some loss; MATTR (.32), function-word ratio (.46) and bigram entropy (.49) are rewritten by the target language. A classifier trained on native target-language responses, which has seen none of these texts, scores .521 (Google) and .531 (LLM) on them on average; an English-trained n-gram model without vocabulary adaptation scores .20 to .28 everywhere but Spanish, Latin-script Turkish included, so the loss is vocabulary, not script.

4.4. Resource Gradient and Convergence

Resource gradient (RQ2). We detect no decline along the resource ranking. The cell-level GLM gives $\beta = -0.038$ log-odds per step (SE 0.042, 95% CI -0.12 to $+0.04$, $p = 0.361$ clustered by prompt; $p = 0.226$ by language, with seven clusters; $p = 0.343$ by both). The estimate is under one accuracy point per step, with no consistent ordering (Hindi exceeds Spanish and Turkish). At 80% power, the minimum detectable effect is about 16 points over the six steps; the interval allows a decline of 17. We cannot conclude that accuracy is unchanged. Joshi's two classes give the same picture (mean .675 for the four Class-5 languages against .627 for the three Class-4 ones), as does the n-gram baseline (Hindi .782 exceeds English .750; GLM $p = 0.380$).

Convergence (RQ4). Models do not meet the planned convergence rule. Centroid distance falls from 3.59 z-units in English to 2.69 in Hindi (Spearman $\rho = -0.96$ against rank, $p = 0.003$; prompt-clustered intervals [3.10, 3.78] against [2.04, 2.86]) and between/within scatter from .41 to .18 ($\rho = -1.00$, $p < 0.001$). But the scatter intervals overlap ([.19, .44] against [.03, .20]), silhouette scores span -0.02 to 0.04 with overlapping intervals, and the rule requires separation on all three. The change centers on Grok: mean distance across its four pairs falls from 4.88 in English to 2.78 in Hindi; the six other pairs average 2.73 in English and 2.63 in Hindi ($\rho = -0.46$, $p = 0.302$). Grok's distinct English register disappears outside English; the other four models are as far apart in Hindi as in English, so neither our result nor La Cava et al.'s [3] supports a general convergence claim.

5. Discussion

In this setting, interpretable attribution works beyond English: the three Class-4 languages average .627 against .675 for the four Class-5 languages, inside the $\pm$9-point intervals, and classifiers transfer across languages after unlabeled standardization. Table 2 gives the practical choice: n-grams when labeled target-language data exist and no explanation is needed, the features when a decision must be explained or must cross languages. Neither method establishes authorship outside the tested candidate set, and translation combined with a paraphrase attack [12] is untested.

Dual use. The features also apply to human text, and human attribution already uses UD-derived features [10], so the release publishes a method rather than a new capability. The translation result is a caution for pseudonymous writers. We release model-generated text only.

Limitations. (i) One genre, persuasive essays with three fixed claims and no lists, so paragraph count and length partly record how each model lays out an argument; both translators keep paragraph boundaries. (ii) One temperature (0.7), no system prompt and one month of model versions; we varied the reading-level instruction but not format instructions or temperature, so a prompt that dictates layout could weaken the strongest single cue. (iii) Length is entangled with style, and the transfer, convergence and translation analyses are not length-controlled; the resource ranking is coarse. (iv) Eight of the 21 features depend on Stanza parses or sentence splitting; parser accuracy varies by language without following the resource order (Hindi and Russian parse best, Turkish and Chinese worst [17]), and four of the five top pooled features are parser-free. (v) The transfer and translation tests share the same 12 prompts across languages, 12 prompts limit the power of prompt-level tests, and prompt fidelity was checked by a model, not by native speakers. (vi) All five subjects are closed APIs; open-weight models, adversarial rewriting and a fine-tuned multilingual encoder baseline [4] were not tested, and judges ran without extended reasoning.

6. Conclusion

In this setting, our 21 interpretable features attribute essays to five frontier LLMs in seven languages at .59 to .72 accuracy: 3 to 22 points below character n-grams within a language, but .49 against .41 across languages with equal unlabeled adaptation. Most of the signal comes from a few features, led by paragraph count and length in the pooled ranking, that pass through translation almost unchanged, and it survives an unseen reading-level instruction and the removal of paragraph features. Next we will vary genre, format instructions and temperature, verify prompts with native speakers, and add a multilingual encoder baseline.

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